ETH zürich

ALMA MATER STUDIORUM
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CachePool: Many-core cluster of customizable, lightweight scalar-vector PEs for irregular L2 data-plane workloads

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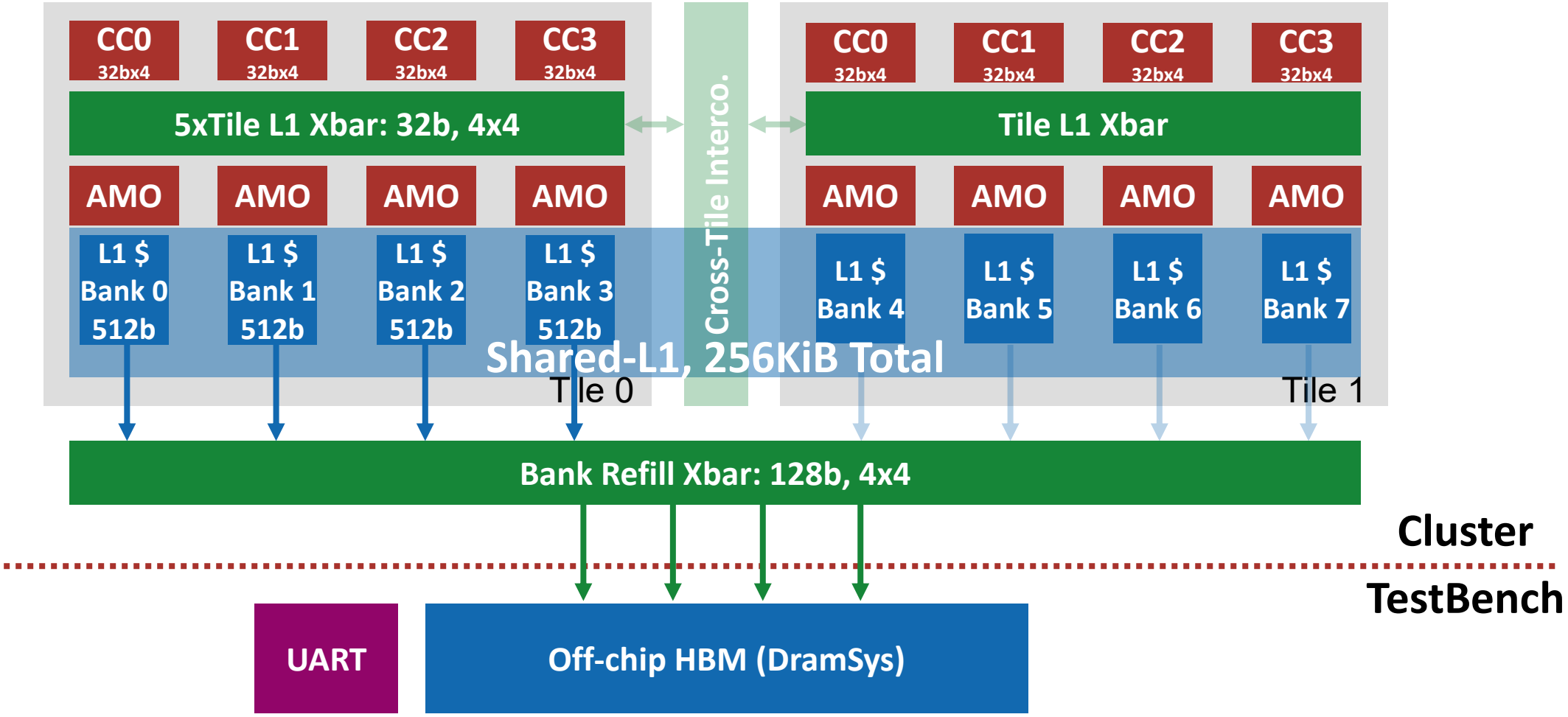
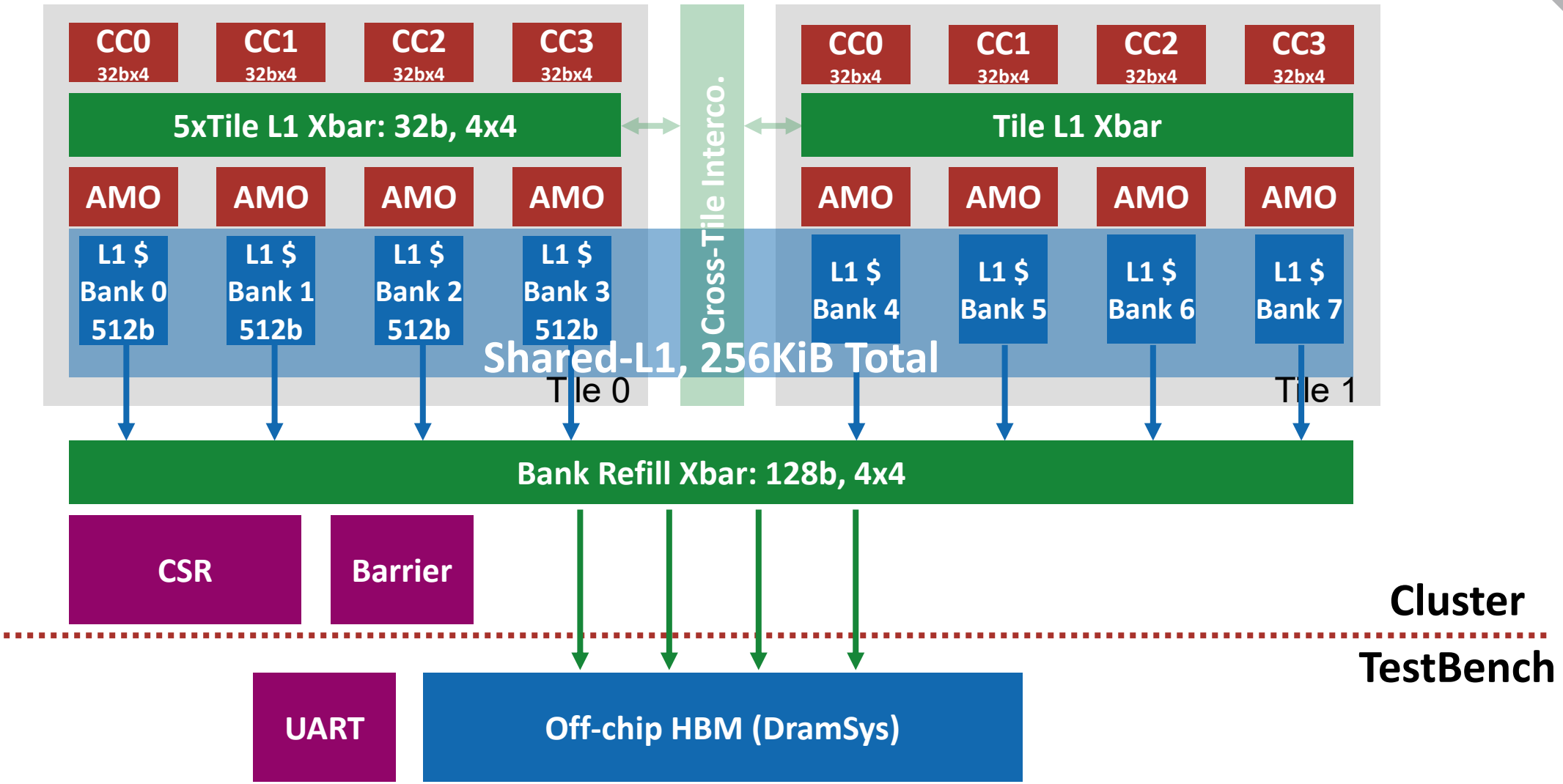
PULP Platform
Open Source Hardware, the way it should be!

@pulp_platform
pulp-platform.org
youtube.com/pulp_platform

Hardware Development (Cluster)



Hardware Development (Cluster)



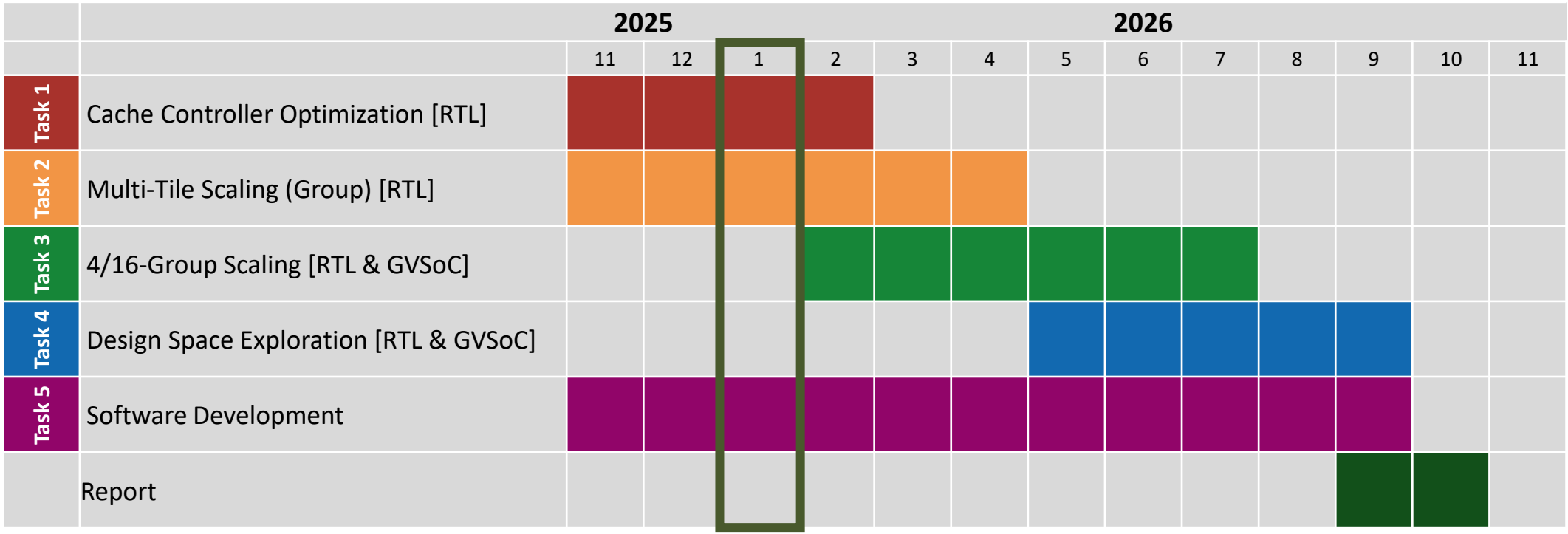
Hardware Development (Cluster)

- **WIP: Multi-Tile Configuration**
  - Now all tiles can be booted and worked on kernels
    - Caches work like private caches for now => needs to debug on the address table
    - Cross-tile interconnection implemented but not yet used
  - Add cluster-level global hardware barrier
    - Two stage: local barrier at tile level, then global synchronization
    - TODO: connect with CSR to make it configurable

Hardware Development (L1 Cache)

- **Add byte / half word aligned / misaligned access support**
  - Snitch → L1 Cache Controller bypass data path
  - Spatz → Coalescer → L1 Cache Controller data path
- **Added support load / store instructions**
  - Scalar load : **lb**, **lh**, lw
  - Scalar store: **sb**, **sh**, sw
  - Vector load: **vle8.v**, **vle16.v**, vle32.v
  - Vector store: **vse8.v**, **vse16.v**, vse32.v
  - Vector indexed load: **vluxei8.v, vloxei8.v; vluxei16.v, vloxei16.v**, vluxei32.v, vloxei32.v
  - Vector indexed store: **vsuxei8.v, vsoxei8.v; vsuxei16.v, vsoxei16.v**, vsuxei32.v, vsoxei32.v
  - Vector strided load: **vlse8.v, vlse16.v**, vlse32.v
  - Vector strided store: **vsse8.v, vsse16.v**, vsse32.v

Timeline


2025
2026
11
12
1
2
3
4
5
6
7
8
9
10
11
Task 1
Cache Controller Optimization [RTL]
Task 2
Multi-Tile Scaling (Group) [RTL]
Task 3
4/16-Group Scaling [RTL & GVSoC]
Task 4
Design Space Exploration [RTL & GVSoC]
Task 5
Software Development
Report

Thank you!